Python Coding Challenge

Please do not share this task and do not publish it online.

The goal of this home task is to assess the candidate's code style, problem-solving methodology and other software engineering skill sets. You have 3 days to submit your solution for the task. If you need extension, please let us know.

Given the ***Expenses.csv*** file containing data representing expenses of different company departments over last a few years broken out on expense types and months. Use this file as a sample of input data for prepared solutions.

Task 1

Prepare a command-line tool that gets input from stdin provided in a format of the ***Expenses.csv*** file and prints results to stdout. The program should aggregate company expenses by a department, a year and a quarter or a month.

There should be the following possibilities of invocation of the program:

a)
`cat Expenses.csv | python3 expenses.py department`
which should return aggregated expenses of a given company department

b)
`cat Expenses.csv | python3 expenses.py department year`
which should return aggregated expenses of a given company department in a specific year

c)
`cat Expenses.csv | python3 expenses.py department year quarter`
which should return aggregated expenses of a given company department in a specific quarter of a year

d)
`cat Expenses.csv | python3 expenses.py department year month`
which should return aggregated expenses of a given company department in a specific month of a year

The output should be a number with the currency of expenses.

Task 2

Create a stateless REST service for the first task (all the data must be provided dynamically). So, the service should receive CSV data via body of a POST request and aggregation parameters as request query parameters.

Make sure the service handles invalid input and parameters properly.

If possible, try to use some Python asynchronous Web framework to build the Web application upon.

The solution should be organized as one Python project.

Task 3

Try to write some tests, at least one, for the prepared solution that could prove it works as expected. You can use any Python test framework that you are familiar with and is widely used in the community.

Task 4 (Optional)

In addition to that you can also propose a database schema to store all data included in the ***Expenses.csv*** file that will allow to query exactly the same information directly from a database.